

Birla Institute of Technology and Science, Pilani

EEE F313/INSTR F313 Analog and Digital VLSI Design

Second Semester 2023-2024

MID-SEMESTER TEST (OPEN BOOK)

Time: 90 min

Max. Marks: 75

Date: 14-03-2024

Q1 (a). It is required to fabricate NMOS and PMOS side by side in order to overcome the drawbacks like crosstalk, latch-up, and mutual coupling effects. [10 Marks]

- Name the IC technology process that you would prefer to fabricate NMOS and PMOS mentioned above.
- With suitable and neat sketches, explain the fabrication steps involved.

(b). Calculate the threshold voltage of a MOS system using $\phi_{ms}=0.88$ V. Consider a p-type silicon substrate at $T=300$ K doped to $N_a=10^{15}$ cm⁻³. Let equivalent oxide charge density, $Q_{ss}=10^{10}$ cm⁻², $t_{ox}=12$ nm=120 Å, and assume the oxide is silicon dioxide with $\xi_{ox}=(3.9 \times 8.85 \times 10^{-14})$ F/cm. [10 Marks]

Q2. For the circuit shown in Figure Q2,

- Calculate the bias current I_D of M_1 . Assume $\mu_n C_{ox} = 100 \mu\text{A/V}^2$ and $V_{TH}=0.4$ V. [5 Marks]
- If the gate voltage increases by 10 mV, what is the change in the drain voltage? [5 Marks]
- Determine the value of W/L that places M_1 at the edge of saturation and calculate the drain voltage change for a 1-mV change at the gate. [5 Marks]
- Calculate the maximum allowable gate voltage such that M_1 must be operated in saturation. [5 Marks]

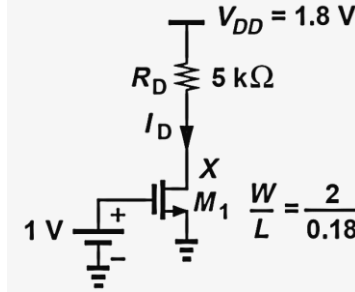


Figure Q2

Q3. For the circuit depicted in Figure Q3,

- Draw the small-signal equivalent circuit. [5 Marks]
- Do M_1 and M_2 appear in parallel or series, justify your answer. [5 Marks]
- Derive the expression for the small-signal voltage gain, $A_V = \left\{ \frac{V_{out}}{V_{in}} \right\}$. [10 Marks]

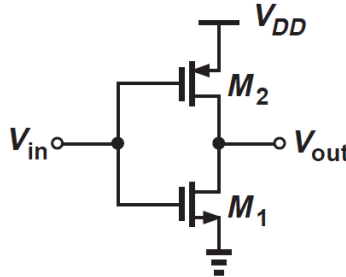


Figure Q3

Q4. A NMOS carries a drain current of 1 mA with $V_{DS} = 0.5$ V while operating in saturation,

- Determine the change in I_D if V_{DS} rises to 1V and $\lambda = 0.1\text{V}^{-1}$. [5 Marks]
- Determine the output impedance of the device? [5 Marks]
- For a MOSFET operating in saturation, how do g_m and $(V_{GS} - V_{TH})$ change if both W/L and I_D are doubled? [5 Marks]